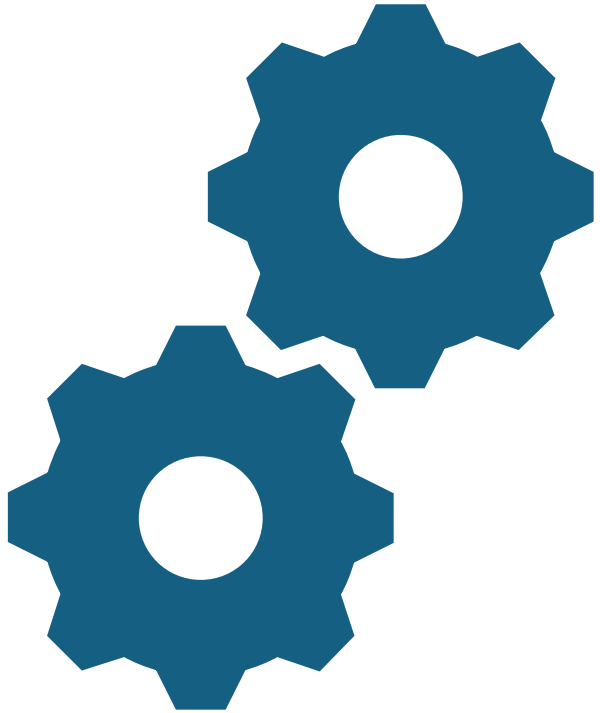
Advanced Deterministic Systems Optimization

ISE 555 Midterm Review



Topics We Have Covered

- The nature of optimization and writing linear programs
- Feasibility, basic solutions, and feasible points
- Descent and optimality
- Writing linear programs in different forms
  - Standard, Canonical Primal, Dual
  - Adding slack and excess, expressing bounds
  - Converting between the forms
- Solving linear programs graphically
- Solving linear programs algebraically
- Solving linear programs with the Simplex Method
  - Finding a starting point
  - Using Tableaus
  - Knowing when to stop and writing the solution
  - Dealing with difficult situations (no good starting basis, infinite solutions)
- Duality of linear programs

The Nature of Optimization and Writing Linear Programs

- Optimization implies a well-stated objective
- Convert the problem statement into variables (things we adjust) and parameters (things that are fixed)
- Add constraints based on practical things like total cost, total availability of resources, total time
- Don’t forget to add any bound constraints on variables
- After solving always look to see if you answered the original question!

Feasibility, Basic Solutions, Feasible Points, and Extreme Points

- These terms all have to do with constraints (both regular and bounds)
- Feasible points meet all the constraints – the set of all of them is the feasible region
- Basic solutions are solutions that meet combinations of constraints
- Extreme points are basic solutions that meet all constraints (they are both basic and feasible) and they are the corners of the boundary of the feasible region

Descent and Optimality

- For linear programs, we showed that optimality can only happen either

  (i) at an extreme point

  or (ii) optimum is unbounded

- From one extreme point we found others by first finding feasible descent directions

- Feasible directions from $x \rightarrow x + \alpha p$:
  - p is in the null space of the constraints for an equality constraint
  - p must meet $a_i^T p \geq 0$ for any **active** constraints of the form $a_i^T x \geq b_i$

- Feasible amount of descent α:
  - We can move any amount in feasible directions for inactive or equality constraints
  - For active inequality constraints, we have a ratio test to find when we cross boundaries of other constraints

Writing linear programs in different forms

Standard Form

$$\begin{aligned} \text{minimize} \quad & z = c^T x \\ \text{subject to} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

where $b \geq 0$

Canonical Primal Form

$$\begin{aligned} \text{minimize} \quad & z = c^T x \\ \text{subject to} \quad & Ax \geq b \\ & x \geq 0 \end{aligned}$$

Dual Form

$$\begin{aligned} \text{maximize} \quad & w = b^T y \\ \text{subject to} \quad & A^T y \leq c \\ & y \geq 0 \end{aligned}$$

Converting to Standard Form

$$\begin{aligned} \text{minimize} \quad & z = c^T x \\ \text{subject to} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

- What to do about maximize?
  - A: Multiple objective by -1
- What if the objective has a constant (ie: $z = 2x_1 + x_2 - 4$)?
  - A: We can ignore it and optimize $z' = 2x_1 + x_2$
- What if a constraint has $b < 0$?
  - A: Multiple the constraint by -1 (note that flips any inequalities)
- What to do about variables that have non-zero bounds (ie: $x_1 \geq 2$)?
  - A: Form $x' = x_1 - 2$ then $x' \geq 0$
- What to do about upper and lower bounds (ie: $x_1 \geq 0$ and $x_1 \leq 7$)?
  - A: Treat the upper bound as a general constraint
- What to do about unbounded variables (ie: x_2 is unbounded)?
  - A: Replace with two new variables x' and x'' with $x_2 = x' - x''$ and $x', x'' \geq 0$
- What about inequality constraints of less than (i.e. $x_1 + x_2 \leq 2$)?
  - A: Add a slack variable s_1 as $x_1 + x_2 + s_1 = 2$ and $s_1 \geq 0$
- What about inequality constraints of greater than (i.e. $3x_1 + 2x_2 \geq -10$)?
  - A: Add an excess variable e_1 as $3x_1 + 2x_2 - e_1 = -10$ and $e_1 \geq 0$

Benefits of Standard Form

- In standard form, a basic solution solves the constraint $Ax = b$
- Feasible directions are now limited to null space calculations
- In standard form, a basic solution is feasible (and hence an extreme point) if it also meets the bound constraint $x \geq 0$
- All extreme points meet this property
- This allowed us to use linear algebra to move between extreme points
- The simplex tableau simplified doing the linear algebra to find neighboring extreme points that lower the objective

Review of the Simplex Method approach

To solve this with the Simplex Method we had three major steps (plus a step 0):

0) Start with a problem in standard form

1) Find an initial basic feasible solution:

   - If we can find a basis that leads to a feasible solution that's great
   - What makes it a basis?
   - What can we do if that fails?
   - Two-phase approach first solves a modified problem with artificial variables a_i and the objective $\min \sigma\ a_i$ The first phase moves from infeasible to feasible objectives by reducing the a_i values The second phase is the "regular simplex"
   - The Big-M method simplifies the two-phase approach by adding $\sum Ma_i$ to the existing objective Then both the movement to the feasible and "regular simplex" are done simultaneously

2) We can use Tableaus to move from one basic feasible solution to a better one

3) We have a stopping criteria when all of the components of the objective row in the Tableau are non-negative

TABLEAU TIPS

- Starting basis must have terms that each exist in only one constraint and not in the objective
- Each basis (and hence each tableau iteration) corresponds to a different extreme point
- The rhs of the objective row should be increasing at each iteration (objective decreasing)
- For a given basis the basis variables are the only ones that are non-zero (all other variables are zero
- If your tableau converges but has a negative variable – you've made an error
- If your tableau hasn't converged but you can't get rhs/x to be positive, then your solution is unbounded

Duality of linear programs

- Dual problems create new variables
- A problem with n variables and m constraints in canonical primal form has m variable and n constraints in dual form
- Dual variables represent marginal gains when the primal variables are quantities in production – thus they are good for resource evaluation
- The *complementary slackness theorem* allowed us to find dual solutions from the canonical primal solution:
  - The theorem says that $x^T\ (c - A^T y) = 0$ for any solutions that solve both the primal and dual problem
  - If any x values are *non-zero*, then the corresponding dual constraint must be met with equality
  - If any canonical primal constraints are *met without equality*, then the corresponding dual variable y must be zero

Solving Dual Problems

- Dual problem form comes from the canonical form (the primal problem)
- To move from standard form to canonical form we need to make the $=$ constraints become $\geq$ constraints
- Approach:
  - Make the objective in standard form (convert to minimization and remove constants)
  - Add extra variables and/or constraints to fix the bounds on the variables
  - DO NOT add any slack or excess variables to regular constraints (ok if you do, just makes more work)
  - For any $\leq$ constraints, multiply by minus one (recall it's ok in canonical form to have $b < 0$)
  - For any natural equality constraints, we replace with two constraints:
    - $a^{\mathrm{T}}x = b$ ➔ $a^{\mathrm{T}}x \leq b$ and $a^{\mathrm{T}}x \geq b$ ➔ $-a^{\mathrm{T}}x \geq -b$ and $a^{\mathrm{T}}x \geq b$
- This is typically fewer variables than standard form (maybe slightly more equations)
- Now form the dual problem:
  - Make the objective in standard form (convert to minimization and remove constants)
  - new variables y
  - transpose the constraint matrix A
  - convert the objective coefficients c^{T} into the constraint constants c
  - convert the constraint constants b into the objective coefficients b^{T}